

ALDDREU AF, Korea, Republic of

WMO: 471870

Lat: 33.200N

Lon: 126.267E

Elev: 89

StdP: 14.65

Time Zone: 9.00 (E09)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.7	30.9	19.5	14.7	33.8	21.4	16.1	34.3	41.0	34.8	38.9	35.7	20.5	350	0.504

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.2	87.8	79.6	86.0	79.1	84.2	78.1	80.7	85.3	79.9	84.0	79.0	82.7	8.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.2	152.0	83.0	78.9	150.3	82.6	77.9	145.4	81.6	44.4	86.0	43.6	84.5	42.8	82.5	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
37.7	32.1	27.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			23.3	82.8	3.6	1.5	20.7	83.9	18.6	84.7	16.6	85.6	14.0	86.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	59.7	40.9	41.4	47.0	55.1	63.0	69.9	77.6	79.9	73.4	64.9	56.1	45.9	(6)
(7)		DBStd	14.23	5.65	6.12	5.59	5.20	3.97	3.29	3.88	3.01	4.24	4.89	6.35	6.22	(7)
(8)		HDD50	853	286	249	125	16	0	0	0	0	0	19	158	(8)	
(9)		HDD65	3282	747	661	559	297	86	4	0	1	63	272	592	(9)	
(10)		CDD50	4390	4	7	31	169	404	596	855	928	701	462	202	31	(10)
(11)		CDD65	1342	0	0	0	0	24	150	390	463	252	59	4	0	(11)
(12)		CDH74	8578	0	0	0	0	27	306	2811	4141	1188	104	1	0	(12)
(13)		CDH80	1853	0	0	0	0	0	14	613	1034	191	1	0	0	(13)
(14)	Wind	WSAvg	11.9	14.0	14.7	13.7	11.8	11.5	9.9	9.1	10.6	9.7	10.6	12.9	14.1	(14)
(15)	Precipitation	PrecAvg	59.9	2.2	2.5	3.9	4.8	5.6	7.6	8.5	10.2	6.6	2.8	3.1	2.0	(15)
(16)		PrecMax	93.7	5.1	6.1	6.9	11.2	12.0	15.6	27.9	23.2	21.8	10.2	10.8	4.8	(16)
(17)		PrecMin	34.2	0.4	0.5	0.6	1.5	2.5	1.8	0.5	3.0	0.5	0.2	0.1	0.3	(17)
(18)		PrecStd	12.8	1.1	1.3	1.6	2.3	2.7	3.3	5.9	5.4	4.7	2.1	2.4	1.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.9	60.5	62.8	70.2	77.2	82.1	91.0	91.0	87.1	79.1	72.1	64.6	(19)
(20)		MCWB	55.3	56.7	55.6	61.9	65.5	70.3	79.6	80.6	78.3	70.9	65.0	60.0	(20)	
(21)		2%	DB	55.4	56.9	60.0	66.6	73.8	79.1	87.6	88.1	84.4	76.9	69.6	61.0	(21)
(22)		MCWB	51.1	53.0	55.2	60.1	64.9	70.0	79.5	79.8	76.6	68.7	63.2	55.7	(22)	
(23)		5%	DB	52.1	53.9	57.6	64.8	71.9	77.2	85.6	86.3	82.3	75.0	67.8	57.6	(23)
(24)		MCWB	47.8	49.7	52.7	59.6	64.2	69.8	79.0	79.4	75.4	67.4	62.3	52.3	(24)	
(25)		10%	DB	49.9	51.4	55.7	62.9	69.9	75.4	83.1	85.4	80.3	72.9	65.2	55.5	(25)
(26)		MCWB	46.1	46.8	50.9	58.2	63.4	69.6	78.0	79.0	74.1	65.8	60.1	50.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.8	57.6	59.3	65.0	70.0	75.8	81.5	82.4	80.5	74.2	67.5	61.1	(27)
(28)		MADB	58.5	59.7	60.7	67.4	73.2	77.4	85.8	86.4	84.3	76.8	69.9	64.2	(28)	
(29)		2%	WB	52.1	54.2	57.1	62.8	68.4	74.2	80.3	81.0	78.9	71.3	65.4	56.4	(29)
(30)		MADB	54.7	56.1	59.4	65.8	71.2	75.8	84.8	85.6	82.2	75.1	68.1	60.5	(30)	
(31)		5%	WB	48.5	50.3	54.4	60.9	66.9	72.7	79.6	80.3	77.7	69.0	63.0	53.4	(31)
(32)		MADB	51.6	53.6	56.9	63.6	69.9	74.8	83.5	84.7	80.9	72.9	66.4	57.6	(32)	
(33)		10%	WB	46.0	46.9	51.7	59.3	65.1	71.2	78.8	79.4	75.8	67.0	60.6	50.8	(33)
(34)		MADB	49.6	50.6	55.3	62.2	68.4	74.0	82.2	83.4	79.0	71.6	64.7	55.1	(34)	
(35)	Mean Daily Temperature Range	MDBR	9.8	10.6	11.6	12.3	11.7	10.2	8.3	9.2	11.3	12.3	11.3	9.7	(35)	
(36)		5% DB	MCDBR	13.4	13.9	13.9	13.8	14.3	13.1	11.5	10.8	13.2	14.5	13.6	13.7	(36)
(37)		MCWBR	11.9	11.6	10.9	10.7	9.4	8.1	6.4	6.4	8.6	10.2	11.0	11.8	(37)	
(38)		5% WB	MCDBR	13.0	13.0	11.9	11.7	10.8	8.9	9.0	9.5	10.6	12.6	12.4	12.9	(38)
(39)		MCWBR	12.6	11.8	11.0	10.4	8.6	6.6	5.7	6.1	7.5	10.3	11.8	12.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.470	0.509	0.558	0.567	0.573	0.601	0.585	0.539	0.496	0.470	0.467	0.464	(40)	
(41)		taud	1.940	1.879	1.779	1.815	1.855	1.877	1.967	2.103	2.161	2.096	2.022	1.959	(41)	
(42)		Ebn at Noon	223	230	230	236	236	229	232	240	245	239	223	216	(42)	
(43)		Edh at Noon	48	56	67	67	65	64	58	50	45	45	44	45	(43)	
(44)	All-Sky Solar Radiation	RadAvg	627	899	1292	1538	1646	1443	1604	1653	1365	1187	809	585	(44)	
(45)		RadStd	75	116	104	129	180	179	224	164	128	118	92	59	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page